

Cross-supplier SoH forecasting for second-life LFP cells using a PyBaMM-trained neural operator

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Second-life reuse of retired lithium iron phosphate (LFP) cells is limited by the cost of predicting future ageing for each cell. Conventional lifetime testing takes months and empirical models often transfer poorly across supplier-specific LFP cell designs. We present a physics-conditioned neural operator that forecasts the future SoH trajectory of a used LFP cell from a short diagnostic characterisation plus the first 50 operating cycles, over an approximately 500-cycle horizon into the deeper second-life regime.

Workflow (Figure 1). Seven anchor cells from three suppliers (CALB, EVE, REPT) are characterised with OCV, HPPC, GITT and RPT (7–9 days per cell, one-time cohort cost) and long-term-cycled at 0.25–0.5 C. Beginning-of-life and five degradation parameters (SEI growth, lithium plating, negative-electrode active-material loss) are identified by differential evolution against measured cycling. PyBaMM [1] SPMe on Prada2013 LFP chemistry [2] then synthesises 490 perturbed degradation trajectories around the anchors; SoH-offset augmentation expands this to 14,096 (context, target) training pairs covering the second-life SoH range. A monotonic θ -conditioned Neural ODE [3] with a learned context encoder combines three inference inputs: a supplier-level θ prior from anchor DE fits, DCIR and initial-capacity features from a ~ 2 -hour characterisation, and a $K=50$ -cycle SoH context window from the target cell.

Results. Evaluation on one cell per supplier that was excluded from neural-operator training (CALB_0029, EVE_0003, REPT_0031) gives SoH RMSEs of 0.95, 0.83 and 0.82 percentage points on the experimentally observed forecast region (Figure 2); mean 0.87 pp, cross-supplier standard deviation 0.06 pp. Comparable errors across suppliers provide preliminary evidence of cross-supplier transfer within the evaluated LFP families. The trained operator predicts a 500-cycle trajectory in **18 ms**, versus ~ 6 s for the equivalent PyBaMM SPMe simulation ($\sim 350\times$ speed-up), enabling per-cell inference at the pace of a field-screening BMS query rather than an overnight compute job.

Limitations. Cross-supplier claim rests on three excluded cells; the 500-cycle forecast horizon is bounded by the 2500-cycle PyBaMM training corpus; and the extrapolation beyond the observed cycling window is not yet experimentally verified. A baseline comparison against linear/exponential fits and against the operator without SoH-offset augmentation is planned next.

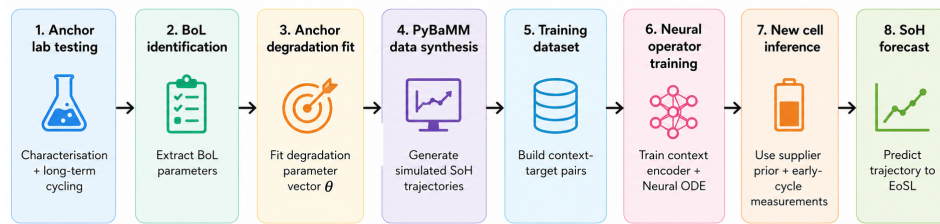


Figure 1: Pipeline: anchor characterisation and DE-fit feed a PyBaMM synthesis corpus; the SciML operator ingests θ , DCIR/capacity features and $K=50$ observed cycles to forecast SoH.

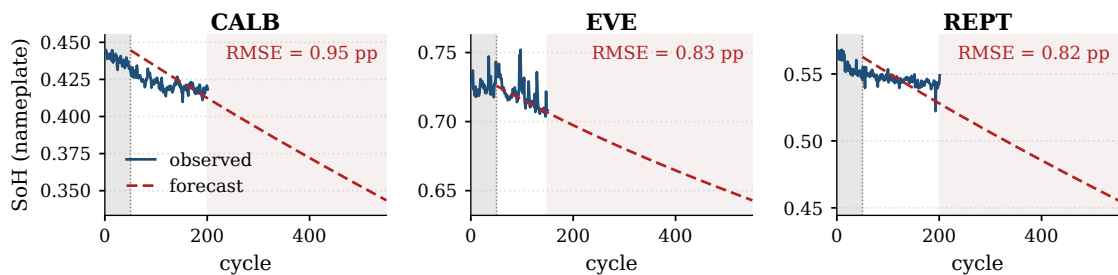


Figure 2: Held-out forecasts (one cell per supplier). Grey band = $K=50$ context; blue = observed; red dashed = forecast; pink band = unverified extrapolation beyond observed window. Per-cell RMSE 0.95, 0.83, 0.82 pp (mean 0.87, std 0.06).